

Initial settings for volume-controlled mechanical ventilation in children*

	Infant (<1 year of age)	Toddler/child (1 to 12 years)	Adolescent (>12 years)
Tidal volume (mL)	5 to 8 mL/kg (healthy lungs)	5 to 8 mL/kg (healthy lungs)	5 to 8 mL/kg (healthy lungs)
	3 to 6 mL/kg (lung protective strategy)	3 to 6 mL/kg (lung protective strategy)	3 to 6 mL/kg (lung protective strategy)
Rate (breaths/minute)	20 to 30	15 to 25	12 to 20
PEEP (cm H₂O)	3 to 8	3 to 8	3 to 8
Pressure support (cm H₂O)[¶]	Minimum 6 to 10	Minimum 6 to 10	Minimum 6 to 10
Peak inspiratory flow (L/minute)	Adjusted to fit desired inspiratory time	Adjusted to fit desired inspiratory time	Adjusted to fit desired inspiratory time
Inspiratory time (seconds) Targeted, based upon changes to inspiratory flow	0.4 to 0.6	0.7 to 0.9	0.9 to 1.2
FiO₂ (%)^Δ	Start with 1.0, rapidly wean to ≤0.6	Start with 1.0, rapidly wean to ≤0.6	Start with 1.0, rapidly wean to ≤0.6
Flow trigger (L/minute)	0.25 to 0.5	0.8 to 2	0.8 to 2
Pressure support cycle	10 to 25% of peak flow rate	10 to 25% of peak flow rate	10 to 25% of peak flow rate

PEEP: positive end-expiratory pressure; FiO₂: fraction of inspired oxygen.

* Consultation with an expert in the mechanical ventilation of children (eg, pediatric intensivist or pediatric anesthesiologist) is strongly encouraged. Regardless of ventilator settings employed, the physician must assess ventilator settings shortly after initiation and frequently thereafter and adjust them as needed to meet oxygenation and ventilation goals as the natural course of the underlying pathophysiology evolves.

¶ Adjust according to endotracheal tube size: 3 to 3.5 mm: 10 cm H₂O; 4 to 4.5 mm: 8 cm H₂O; ≥5 mm: 6 cm H₂O.

Δ Wean to 0.6 or below to maintain arterial oxygen tension (PaO₂) 60 to 80, oxygen saturation (SpO₂) 92 to 97% when required PEEP is <10 cm H₂O; if required PEEP is ≥10 cm H₂O, then targeted oxygen saturation may be reduced to 88 to 92% for those patients with pediatric acute respiratory distress syndrome (PARDS).